MP1 – Research methodology

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Scheduling

Balanced theory-practice program (12h + 4h) :



- **4h – Research methodology** – *JP. Pernot (October 8th, 2025)*
- **4h – Information skills and bibliographic resources**
  C. Wahnoun (October 10th, 2025) Bring your laptop with Zotéro installed
- **1h – Presentation PJR subjects** – *All (October 13rd, 2025)*
- **1h – General rules on data protection**
  Intellectual property department – A. Bouderbala (October 15th, 2025)
- **Kick-off PJR** – *All (October 20th, 2025)*
- **4h – State-of-the-art analysis/presentation**
  JP. Pernot (November 10th, 2025)
- **2 x 2h – Check intermediate milestones PJR (research design canvas, state-of-the-art, validation approaches, etc.)**
  L. Roucoules & JP. Pernot (28/11/25 et 12/12/25)

Written exam (December, 2025)

Final mark MP1 = 25% STAR oral presentation + 75% written exam

Aims

To explain what research is, and what it is not,

To introduce the objectives of research

To outline the different types of research

To discuss the research process

To introduce the concept of “Research Problem”

To warn of common mistakes

To understand the literature review basics

To discuss ethical issues



Overview

Fundamentals

Starting your own project

Literature review basics

Ethical issues



References

References on research methodology :

- *Research methods : the basics*, N. Walliman, Routledge editor, 2010.
- *Research Project : Designing, Planning, and Getting Started*, N. Walliman, Oxford Brookes University, UK, Sage, 2019.
- *Planification d'un projet de recherche – MTR801*, F. Garnier, École de Technologie Supérieure de Montréal, Canada, 2019.
- *How to Write a Thesis*, R. Murray, Open University Press, 2002.
- *Practical Research,* P. Leedy and J. Ellis Ormrod, 12th Edition, Pearson, 2018.

References on literature review :

- *The literature review : A step-by-step guide for students.* D. Ridley, Sage study skills, 2008.

References on ethical issues :

- *The Student's Guide to Research Ethics.* P. Oliver, Maidenhead: Open University Press, 2003.



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FUNDAMENTALS

What is research ?

Scientific method

Objectives of research

Types of research

The research process

Method vs. Methodology

What is research ?

Received ideas (1/2)

Research does not address abstract and theoretical topics that no one understands and completely disconnected from our daily lives !

It is an activity that is motivated by our need to satisfy our natural curiosity and our wish to make sense of the world around us.



Research is not a mere gathering of facts or information

This kind of activity may more accurately be called **'collection of information'**, and can be carried out in a systematic and in-depth way.

It certainly can be seen as an important part of research.

Ex: The result of the measurements is that the speed of the blood cells in the blood is Xm/s. That's all.



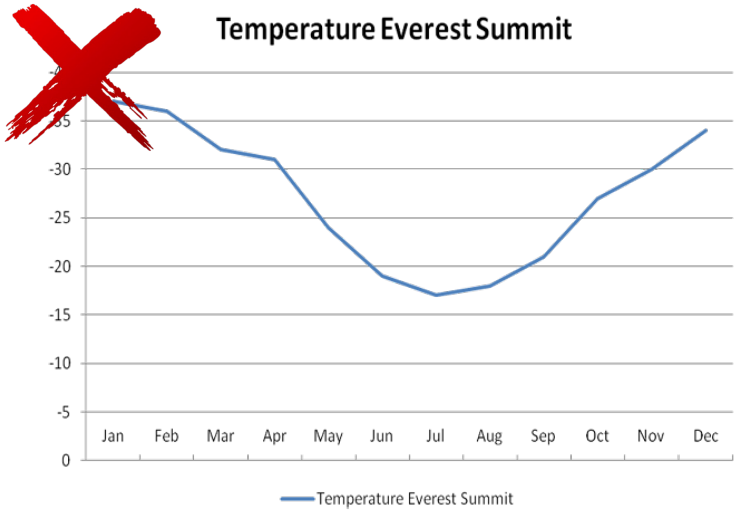
What is research ?

Received ideas (2/2)

Research is not a simple reproduction and presentation of facts

One might call this form of activity **‘assembly of information’**. One vital ingredient of the research process is missing : **the interpretation of the information**. This is, as with the collection of information, an important component of research, but not its entirety.

Ex: Evolution of the Everest summit temperature



Research is not a slogan to attract attention !

There must be evidence of **a research process and methodology**.

Ex: Advertising for toothpaste



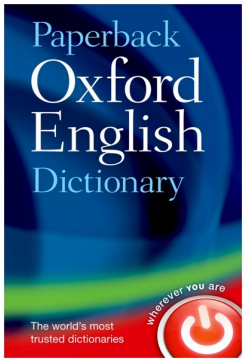
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What is research ?

Tentative definitions

Oxford Encyclopaedic English Dictionary (OEED, 1991)



- The **systematic investigation** into the study of materials, sources etc. in order to establish facts and reach new conclusions.
- An attempt **to discover new** or collate old facts etc. by the **scientific study** of a subject or by a course of critical investigation.

Leedy, 1989

Research is a procedure by which we attempt to find systematically, and with the support of demonstrable fact, **the answer to a question or the resolution of a problem**.

Dominowski, 1980

Research is a **fact-finding activity**.

Kerlinger, 1970

The systematic, controlled, empirical and critical **investigation of hypothetical propositions about presumed relations among natural phenomena**.

Definitions differ in terms of emphasis, scope and means !

What is research ?

More basic and ancient means : Experience and Reasoning (1/2)

Experience (Walliman, 2001)

- **It results in knowledge and understanding** gained either individually or as a group or society, or shared by experts or leaders, **through day-to-day living** and **trial-and-error learning**.



© Brian Crane.

- **Knowledge gained from experience forms an essential aid** to our understanding and activities in everyday life.
- It can be **a valuable starting point for systematic research**, and may provide lots of questions to be investigated and ideas to be tested.
- However, it does have **severe limitations** as a means of methodically and reliably extending knowledge and understanding of the world.

What is research ?

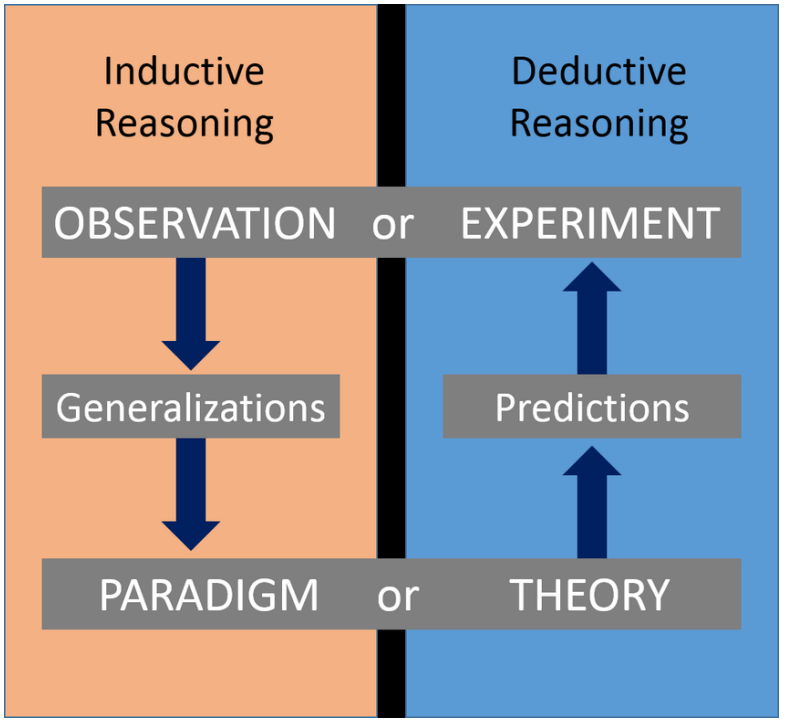
More basic and ancient means : Experience and Reasoning (2/2)

Reasoning (Walliman, 2001)

- It is a method of **coming to conclusions by the use of logical arguments**.
- There are two basic forms of argument :
  - **Deductive reasoning** (from general statements to a specific conclusion).

    Ex : If all organisms are made of cells, and humans are organisms, then humans are made of cells.
  - **Inductive reasoning** (from specific observations to general conclusion).

    Ex : He always leaves for school at 7:00 a.m. and he is always on time. He assumes, then, that if he leaves at 7:00 a.m. for school today, he will be on time.



The combination of experience with <u>both</u> deductive and inductive reasoning is the foundation of modern SCIENTIFIC RESEARCH

What is research ?

Characteristics of research

Three characteristics of research can be seen to distinguish it from gaining knowledge either purely by experience or by reasoning (Walliman, 2001) :

- Gaining experience is an uncontrolled and haphazard activity, while **research is systematic and controlled**.
- Reasoning can operate in an abstract world, divorced from reality, while **research is empirical** and **turns to experience** and the world around us **for validation**.
- Unlike experience and reasoning, **research aims to be self-correcting**. The process of research involves **rigorously testing the results obtained**, and methods and results are open to public scrutiny and criticism.

Research is a combination of both experience and reasoning and must be regarded as the most successful approach to the discovery of truth (Cohen and Manion, 1994).

It is usually assumed that such a systematic research makes use of the rigorous and questioning techniques of scientific enquiry. This form of enquiry is called SCIENTIFIC METHOD.

Scientific method

Assumptions (1/2)

There are 5 major assumptions underlying scientific method (Cohen et al., 1994)

(1) Order :

- Belief that there is **some kind of order in the universe** and that it is **possible to gain some understanding of it**.
- Linked with the **idea of determinism**, the assumption that **events have causes**, and that the **links between events and causes can be revealed**.
- However, it is commonly admitted that due to imperfect knowledge, **predictions of varying levels of probability** often result.

(2) External reality :

- There must be **an agreement between people that external reality exists**, and that people recognize the same reality, a public or shared reality.

(3) Reliability :

- To avoid questioning at every single stage, some credence must be given to the power of human memory, perception and intellect **to provide reliable knowledge**.

Scientific method

Assumptions (2/2)

(4) Parsimony :

- Phenomena should be explained in **as economic a manner as possible**.
- Needless complexity is avoided, and scientists aim to achieve **the most elegant and simple theories**.

(5) Generality :

- There can be valid relationships between the particular cases investigated by the researcher and **the general situation** in the world at large.
- However, **this assumption can be questionable in some sciences** with a large number of unknowns (e.g. sociology).



Some of these hypotheses have raised, and continue to raise, philosophic debates over the years…

Scientific method

Key characteristics of the scientific method (1/3)

Research using the scientific method displays **6 characteristics which distinguish it from other methods** of enquiry (Walliman, 2001) :

(1) It is generated by a question : A questioning mind is the precondition for research. Why, how, what, when… ? Such a question is **often referred to as the research problem**.



(2) It necessitates clarification of a goal : Without **a clear statement of the objectives** and what is intended to be done, the research cannot be successful.



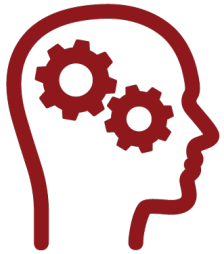
(3) It entails a specific programme of work : Research needs **to be carefully planned** in order to achieve its objectives and reach conclusions.



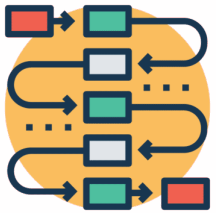
Scientific method

Key characteristics of the scientific method (2/3)

(4) It is aimed at increasing understanding by interpreting facts or ideas and reaching some conclusions : The significance of facts or ideas depends on the way in which the intellect can extract meaning from them.



(5) It requires reasoned argument to support conclusions : In order to communicate an ordered sequence of ideas, a **clear logical argument is required**.



(6) It is reiterative in its activities : Advances are based on previous knowledge, which, in turn, is expanded by the advances. Resolution of problems **often gives rise to further problems** which need resolving.

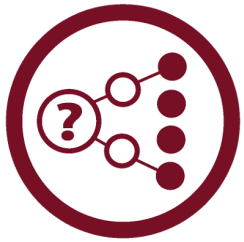


Scientific method

Key characteristics of the scientific method (3/3)

In addition,

Research often **divides the principal question or problem into more practicable sub-questions or problems**, enabling them to be practically investigated.



Research is tentatively **guided by assertions called hypotheses**. Testing these hypotheses provides a direction for exploration.



Research requires **measurable data** in attempting to answer the question which initiated the research.



Objectives of research

Singly or in combination



The main overriding objective must be that of **gaining useful or interesting knowledge.**

In addition, there exists multiple objectives (Reynolds, 1971) :

Categorization : It involves **forming a typology of objects, events or concepts**. This can be useful in explaining what 'things' belong together and how.

Explanation : Being able **to understand and to provide explanation of particular phenomena** is a common objective in many forms of research.

Prediction : If it is possible to explain a phenomenon, then **future events related to it can be predicted**.

Control : A good level of understanding of a phenomenon might lead to **the possibility of finding a way to control it**.

Evaluation : It is about **judging the quality** (measured in an absolute way, or on a comparative basis) of objects or events.

Types of research

Quantitative vs. qualitative researches

Researches can be classified according to whether they correspond to a :

Quantitative research :

- Traditional or experimental research **with well-defined variables**.
- Collected information is appraised **by counting and assessing numbers**.

Qualitative research :

- **Interpretative research** used for the study of complex phenomena.
- Collected information is appraised **by measuring and evaluating qualities**.

More precisely, they differ in the following characteristics :

(1) The purpose of the research :

- **Quantitative** : Explain and predict / Confirm and validate / Test theories
- **Qualitative** : Describe and explain / Explore and interpret / Build theories

Types of research

Quantitative vs. qualitative researches

(2) The nature of the research process :

- **Quantitative** : Selective / Known variables / Guidelines established / Predetermined methods / Free of context / Detached view
- **Qualitative** : Holistic = interested by the object or phenomena a whole / Unknown variables / Flexible guidelines / Emerging methods / Context-related Personal view

(3) The type of data collection :

- **Quantitative** : Digital data / Large sample / Standardised instruments
- **Qualitative** : Text and/or image-based data / Small sample / Vaguely structured or non-standardized observations and interviews

(4) Data analysis :

- **Quantitative** : Statistical analysis / Objectivity / Deductive reasoning
- **Qualitative** : Search for themes and categories / Subjectivity and recognized bias allowed / Inductive reasoning

(5) Results and communication :

- **Quantitative** : Numbers / Statistics, aggregated data / Passive form, scientific style
- **Qualitative** : Words / Narrative, quotations / Active form, personal literary style

Types of research

Major types (1/3)

Historical research : Systematic and objective location, evaluation and synthesis of evidence **to establish facts and draw conclusions about past events** (Borg, 1963). This type of research should pay attention to : (i) the analytical rigour, (ii) the authenticity of the contents, (iii) the accuracy of the statements, etc.



Ex : The literature review, that is part of the research process, is in some sense a kind of historical research.

Comparative research : Often used together with historical research. Based on the principle that it is often **easier to understand phenomena when they are compared** with similar phenomena from another time or place.



Descriptive research : It relies on observation as **a means of collecting data**. It attempts to examine situations in order to establish what is the norm, i.e. what can be predicted to happen again under the same circumstances. As this type of research is based on human observations, **a particular attention has to be paid to avoid distorting the data and inserting bias.**



Types of research

Major types (2/3)

Correlation research : It measures the **level of relationship between two phenomena**. The information are represented by **numbers** analysed using statistics for instance.



Experimental research :

- It deals with the **cause-and-effect phenomenon** as part of the research process.
- Cohen and Manion (1994): An experiment involves **making a change in the value of one variable** (called the independent variable) and **observing the effect of that change on another variable** (called the dependent variable).
- After the experiment, data must be processed and analysed, and experimental findings must be formulated and interpreted.



Evaluation research : This is a descriptive type of research designed **to deal with complex social issues**. The evaluation outcomes are not intended to represent 'the way things really are, or how they work', but **present the meaningful constructions** which the individual actors or groups of actors create in order to make sense of the situations in which they find themselves (Guba and Lincoln, 1989).



Types of research

Major types (3/3)

Action research :

- Related to experimental research, but carried out **in the real world** rather than in the context of a closed experimental system.
- A small scale **intervention in the functioning** of the real world and a close **examination of the effects** of such an intervention (Cohen and Manion, 1994).

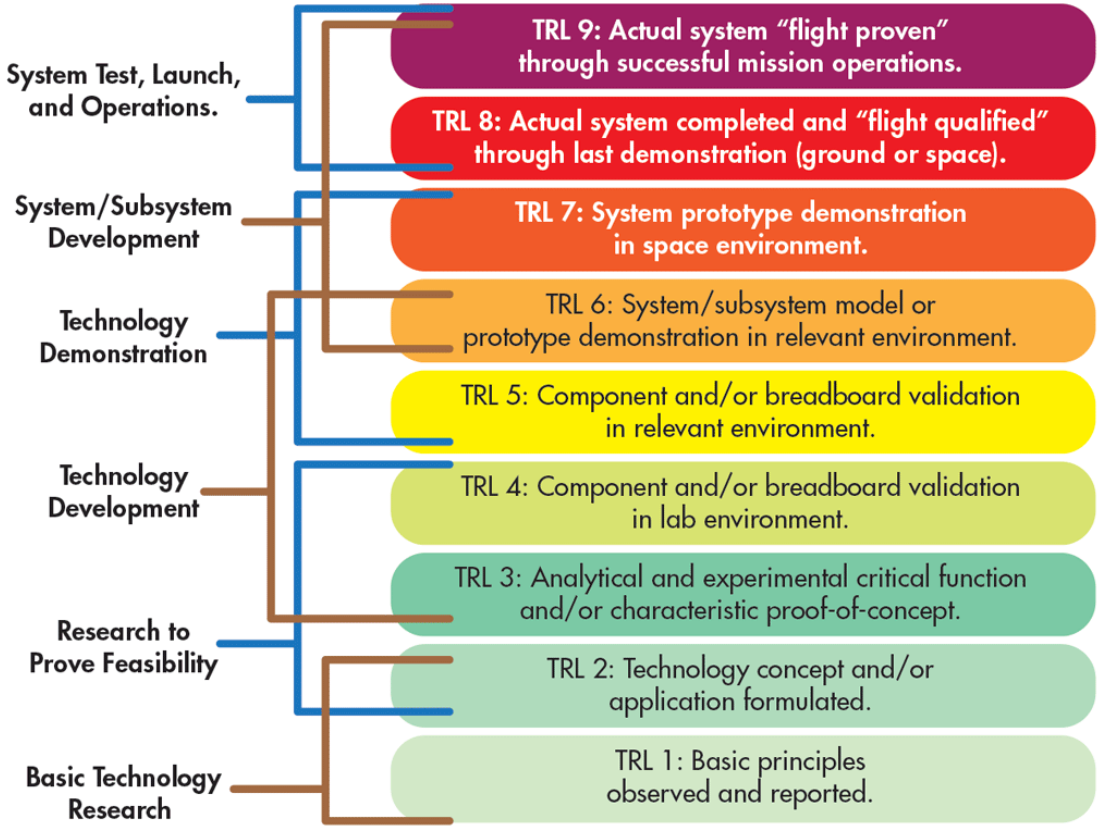


The research project may exploit several of the above mentioned types of research, and **it is usually not restricted to one type**.

Types of research

Technology Readiness Levels

Technology Readiness Levels (TRL by NASA in the 1970s) : It is a method **to estimate the maturity of technologies** during the acquisition phase of a program. The use of TRLs enables **consistent, uniform discussions of technical maturity** across different types of technology.



When undergoing a research project, it is important to understand **where you are on the TRL scale** !

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The research process

Circular process

A rather circular process where :

- Research theory plays a central role
- You can enter at different stage
- The results and conclusions **often raise new questioning** that require **a new research process** to be initiated

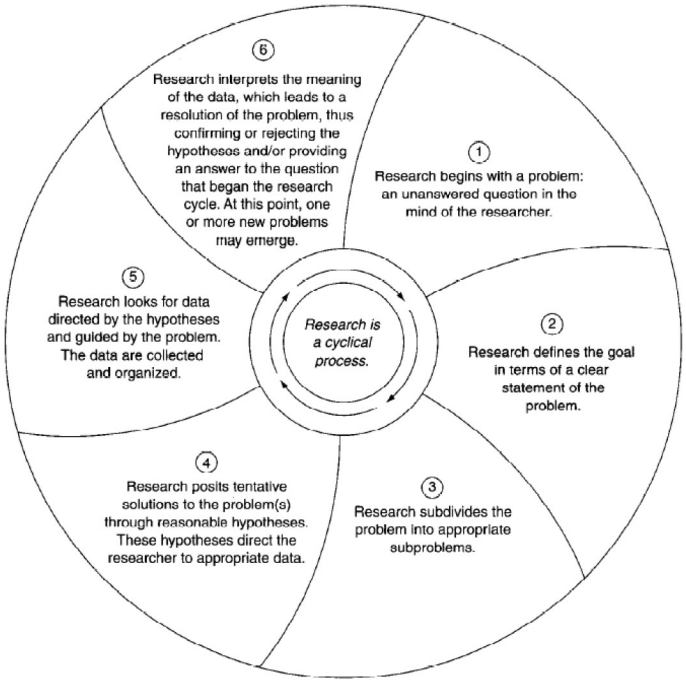


This very minimalist and logical representation must not overshadow the **complexity inherent in any research process** (questioning, dead-end, step back, etc.).

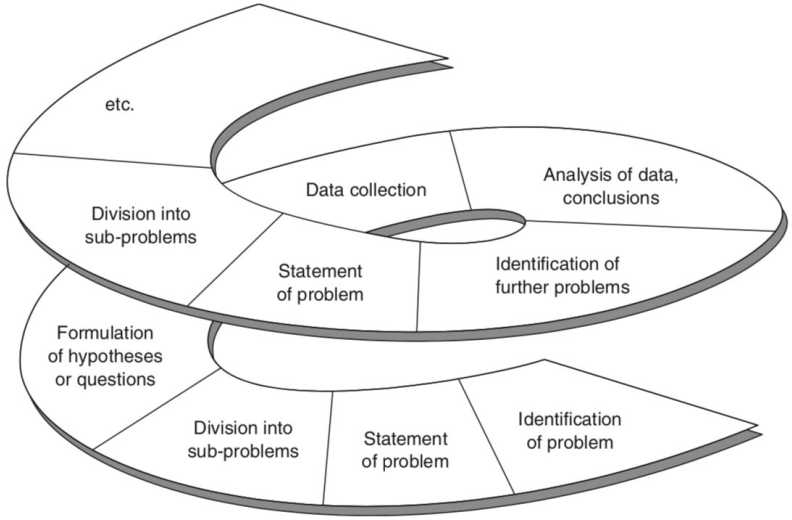
The research process
Cyclical process

A cyclical process where :

- Results and conclusions often raise **new questioning that require a new research process to be initiated**.
- Research is **not a one-time act**, not static, not self-contained, and it does not end in itself.



Cyclical process (Leedy, 1989)



Spiral diagram (Walliman, 2001)

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The research process

Successive widening and narrowing of knowledge bands

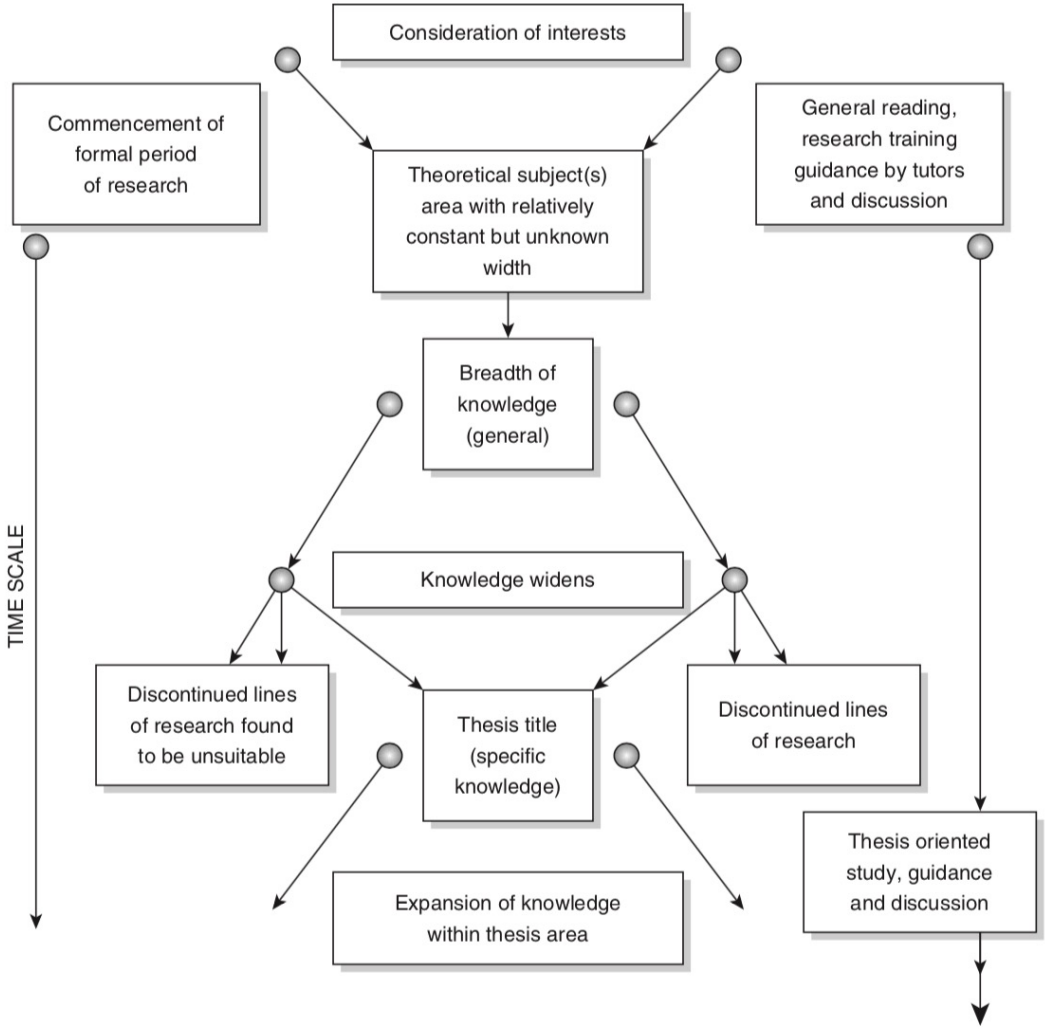
The research process involves **successive widening and narrowing** of knowledge bands (Newman, 1989)

Widening





Narrowing



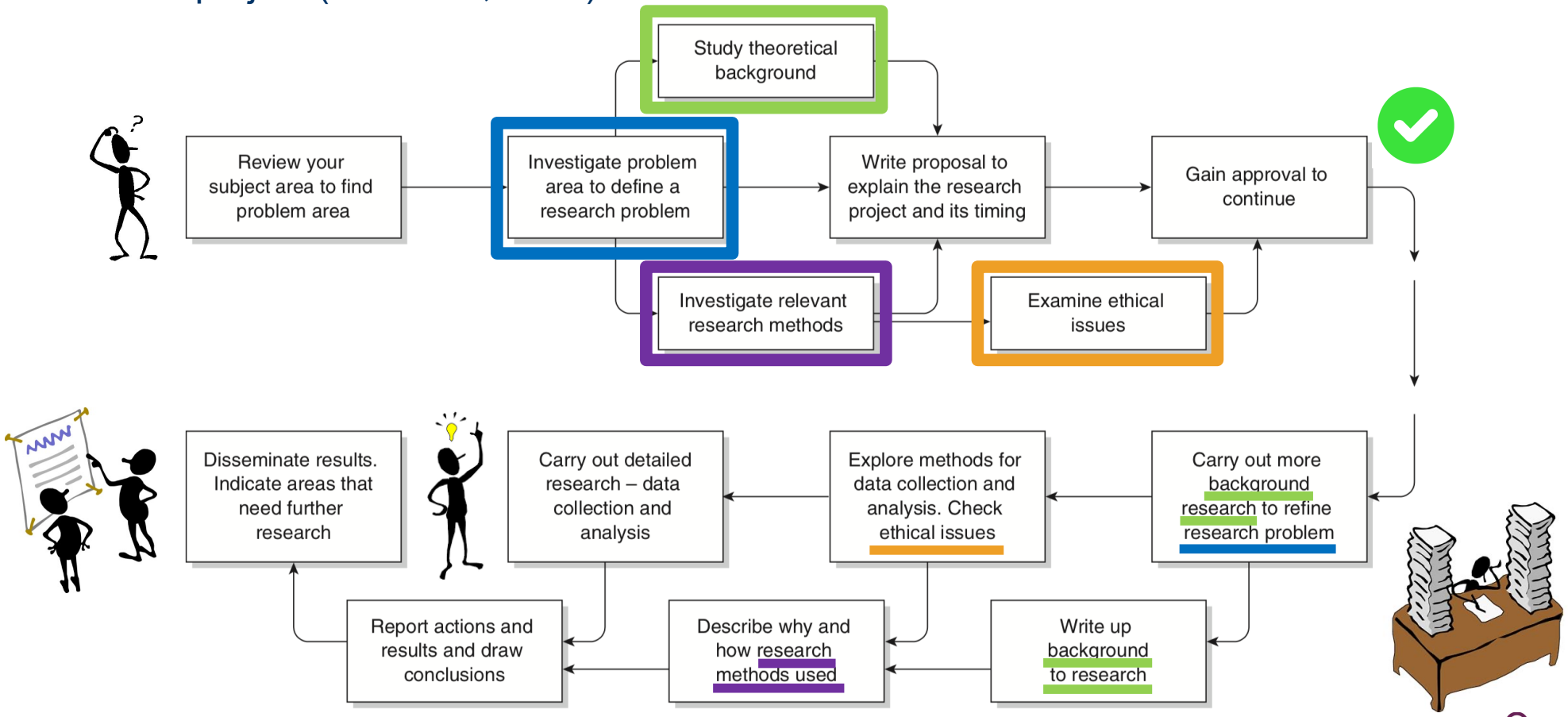
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The research process

A series of interrelated stages

The research process can be seen as **a series of stages that are interrelated and are sometimes revisited in an iterative fashion** during the project (Walliman, 2001)



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Four desirable characteristics of the research findings (Reynolds, 1971) :

- **Abstractness :** Findings should be **independent from a specific time and place**. They are useful **if they can be applied in other situations**, and can lead to the development of general theories.
- **Intersubjectivity :**
  - **Intersubjectivity of meaning :** There must be an **agreement between people about meaning**. It requires precise definition of concepts used in statements.
  - **Intersubjectivity at the logical level :** There must be an agreement, at the logical level, between people on the use of the various statements forming the research findings. It requires the **use of commonly adopted logical systems** (e.g. mathematics, statistics)
- **Empirical relevance :** It is a measure of the **correspondence between a particular theory** and what is taken to be **objective empirical data** resulting from observation and experimentation. This measure must be as great as possible.

Method vs. methodology

Understanding the difference

Method : It is simply **a tool used to answer your research questions**.



- To collect your data, you may use : interview, survey, usability study, etc.
- To analyse your data, you may use : statistics, artificial intelligence, etc.
- To simulate a behaviour, you may use : Finite Element analysis, etc.
- …

Methodology : It describes the **general research strategy** that outlines the way in which research is to be undertaken (Howell 2013).

Example of possible methodology : First, the participants will be identified. Then, interviews will be conducted using questionnaires and video recording. Data will be processed and used as inputs of a neural network so as to get a behavioural model. The resulting model will be tested on other data set so as to validate its efficiency to work on unknown data.





Within **one methodology**, you can apply **several different methods** to support or reject the research hypothesis.

STARTING YOUR PROJECT

Finding and defining a research problem

Some common mistakes

Aids to locating and analysing problems

Research problem definition

The sub-problems

Research design canvas

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Finding and defining a research problem

Heart of the research project

The research problem is the heart of the research project



Within a context and for a particular subject, the research problem arises from :

- A **question** that demands an **answer**
- A **need** that requires a **resolution**
- An **enigma** that seeks a **solution**
- A **gap in knowledge,** an **unresolved controversy**, an **unfulfilled need**

Its formalization requires :

- A good **awareness of current issues** in the subject and context
- An accurate identification of the **core research question**
- A clear **statement of the problem** expressed with **great precision** and **subdivided into sub-problems** easier to manage

The answer/solution to the research problem forms the scientific contribution or findings that must also be clearly made explicit.

Finding and defining a research problem

Features of a suitable research problem (1/2)

It should be of great interest to you. Remember **you will spend several months/years** investigating the problem !

The problem should be significant :

- It is not stimulating to spend time and effort investigating an insignificant problem or repeating work that has already been done elsewhere.
- Of course, this feature has to be weighted, and **the same will not be expected of a Master student as it would of a PhD student**.

The problem should be clearly delineated :

- Consider the time you have to complete the work, and the depth to which the problem will be addressed.
- You can cover a wide field only superficially, and the more you restrict the field, the more detailed the study can be.
- Of course, the problem tackled by a Master student will be more circumscribed than the one of a PhD student.

Finding and defining a research problem

Features of a suitable research problem (2/2)

You should be able to obtain the information required :

- You cannot carry out research **if you fail to collect the relevant information** needed to tackle your problem.
- Make sure **you have access to documents or other resources**, and/or that you can count on the cooperation of individuals or organizations essential to your research.

You should be able to draw conclusions related to the problem :

- The purpose of asking a question is **to find an answer**.
- The problem should be **one to which the research can offer some solution**, or at least the elimination of some false 'solutions'.
- Remember that concluding that what you initially thought was possible is finally not possible, is a contribution.

You should be able to state the problem clearly and concisely. A precise, well thought out and **fully articulated sentence, understandable by anyone**, should normally clearly be able to explain just what the research problem or question is.

Common mistakes

To be avoided (1/2)

Making the choice of a problem an excuse to fill in gaps in your own knowledge :

- The purpose of research is **not just to gain new personal knowledge**, but to make a wider contribution to public knowledge.
- Identifying a research problem requires an additional effort to **find one that requires data to be analysed and conclusions to be drawn** which are of wider interest.
- Of course, new knowledge will be gained during the research activity, but the research process cannot be restricted to this.

Formulating a problem that involves merely a comparison of two or more data sets :

- Without any effort to reveal something new from the comparison, there is no research activity.
- The problem should **clearly state the objectives behind making the comparison**.
- Of course, during the research process, you may have to compare data sets, but the research problem **cannot be restricted to a simple comparison**.

Common mistakes

To be avoided (2/2)

Setting a problem in terms of finding the degree of correlation between two data sets :

- Comparing two data sets to reveal an apparent link between them (e.g. the average age of marriage and the size of families) might be interesting, but the result is only a number, and does not reveal a causal connection.
- This number, or coefficient of correlation, reveals nothing about the nature of the link, and invites **the question – so what ?**

Devising a problem to which the answer can be only yes or no :

- A yes-or-no solution to a problem **does not bring much information** to understand why things are as they are and how they work.
- It avoids the search for the reasons **why yes or no is the answer**, and the implications which the answer has.
- Of course, the answer to a particular research question can be a yes or no, but **it should not be restricted to this**. Such a statement must be argued, discussed so as to clearly understand the scientific reasoning behind it, the hypotheses, the limitations, etc.

Aids to locating and analysing problems

Process for research problem formulation

Booth et al. (1995) suggested a four-steps process to come out with a clear formulation of the research problem :

Find an interest in a broad subject area (problem area) :

- Remember that you will spend months on the subject that has to be of interest for you.
- The problem area **can be broad at the beginning** : reverse engineering of mechanical parts, interoperability between Cyber-Physical Systems, smart manufacturing, etc.

Narrow the interest to a plausible topic :

- Narrow down the scope of the idea or problem until it becomes a **highly specific research problem.**
- This narrowing process will require **a lot of background reading.**



Narrowing process

Question the topic from several points of view

Define a rationale for your project :

- What is your motivation for doing the research?
- What relevant interest, experience or expertise do you bring to the subject ?
- What are you going to produce ?

Aids to locating and analysing problems

Initial literature review and problem area definition

The objective of the initial literature review is to discover relevant material published in the chosen field and to look for a suitable problem area.

At this stage it is important to :

- Get **an overview of the subject**, rather than knowledge in depth.
- Be aware of **the context** within which the research work is to be carried out, as the **literature review strongly relies on the context**.

  Example: segmentation algorithms developed in the 3D modelling context differs from the ones developed in the signal analysis or image processing contexts, even though connections may exist between different fields.
- **Familiarize yourself** with the previously established and widely accepted statements and results.

This stage is more or less time-consuming depending on the level of expertise one already have in the domain.

Aids to locating and analysing problems

Research problem definition

This stage requires a curious mind, an eye for inconsistencies and inadequacies in current theories, and a certain imagination.

The research problem definition can be broken down (Walliman, 2001) :

- Ask a **first simple question**. It is likely that the problem is too broad if you can state it in less than half a dozen words.
- **Additional questions** can be posed and can help delineating the research problem by means of where, what, why, which, when, who questions.
- Try **to order the questions** so as to test various logical progression from sentence to sentence.
- Define the **research question** in a more elegant and synthetic way.



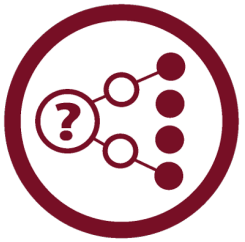
This stage can be iterative as the literature review may reveal new questions themselves requiring new literature reviews and so on**.**

Remember (Newman, 1989) who defined the research process as a set of widening and narrowing steps.

Aids to locating and analysing problems

The sub-problems

Most research problems are difficult, or even impossible, to solve without breaking them down into smaller problems.



The sub-problems should delineate the scope of the work and, taken together, should define the overall problem to be addressed as summarized in the research question.

Booth et al. (1995) elaborated 4 questions to help decomposing the research problem into sub-problems :

- What are the parts of your subject and what broader set is it part of ?
- What is its history and what broader history is it part of ?
- What kind of categories can you find there, and what are the main categories of things it belongs to ?
- What's good about it ? What can you do with it ?

The questions sketched and logically sequenced to define the research problem can help decomposing it into sub-problems.

→ Each sub-problem requires **a more focused literature review**.

Aids to locating and analysing problems

Sub-problem contextualization

Hypotheses must be clearly stated :

- A hypothesis is intelligent and informed conjecture of a possible solution to the research problem. **It shall not be proved !**
- Hypotheses are to be defined per sub-problem.
- A **null hypothesis** can be easier to manage : instead of showing that the working hypothesis is supported, it is shown that the opposite hypothesis is not supported.

Sub-problem must be delimitated :

- The research problem integrates what the researcher has planned to do.
- The **delimitations clearly state what is not supposed to be done** by the researcher, in order to avoid wasting time.

Terms must be defined in your context, especially the technical ones that may be context-dependent and have several interpretations/definitions

Suppositions must be identified as they often are implicit (≠ hypotheses)

Examples : The algorithm will work in 2D space, the input images will be provided by means of the jpeg format, etc.

Research design canvas

Tool for high-quality research

Name:

Title of Thesis:

Date / Version:

Research Design Canvas

Literatures

What bodies of literature or areas of research will be used?
What are the key papers / studies in each literature?
Will each literature play an equally important role in research?
Are the literatures from different disciplines / fields?



Observations & Arguments

What themes / issues are evident in the literatures?
What gap(s) / problem(s) are evident in the literatures?
What are the costs or risks of not addressing the gap(s) / problem(s) for researchers in the literature?



Research Questions

What are the research questions?
What is the value of investigating them?
Are there any tentative hypotheses?



Contributions

What are the potential contributions of the research?
Who are the key stakeholders and what benefits will the research provide to them?



Sample / Context

What sampling methods and sample will be used?
Will the context of the research play an important role?
If so, what are the distinctive characteristics of the context?



Methodology / Design

What is the overall design or methodology for the study?
What are the key principles / objectives of the methodology?
What are the quality criteria for evaluating the design or methodology?



Methods

What methods of data collection and analysis will be utilised?
What are the key features or strengths of each method?
Will the methods be used in a particular sequence?
How do the methods compliment each other?



Theory

Will theory be an important part of your research?
What role will theory play in your project?
What theoretical ideas / concepts / model / framework do you intend to use or develop
What does theory mean or what constitutes theory in your discipline or field?



Problem / Phenomenon

Does a real-world problem(s) motivate the project?
What are the main phenomenon and specific sub-phenomena being investigated?



Assumptions / Paradigm

Is the research based upon particular philosophical assumptions?
Does the research operate within a specific paradigm?



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LITERATURE REVIEW BASICS

Aims of the literature review

Loads of information

Types of sources

Doing a literature review

Presenting your analysis

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Aims of the literature review

What a literature review should be, and what it must not be

The objective of the literature review is to draw on all available sources to :

- Track down **the latest knowledge and findings**. Understanding the ‘How ?” is important as it can give ideas on the preferential methods to solve a particular question.
- **Assess it** for relevance, quality, controversy and gaps
- Identify **where additional research is required**, to try to resolve a controversy or to fill a gap. Understanding the “How ?” is important as a problem can be solved in several ways.

The literature review helps :

- Being **more confidante** on the research problem in the problem area
- Finding **new information sources** that might have been missed, and other researchers working on similar topics
- Identifying **existing methods and approaches** : existing methods and tools, existing approaches and methodologies to solve problems similar to yours

The literature review is not a summary of existing approaches !

The literature review is not a one-time step :

- The researcher often has to come back to this step when moving on in his/her research
- **It never stops !** Knowledge evolves over time.

Loads of information

Quality rather than quantity (1/2)

There is a huge amount of information from different sources around the world (not exhaustive list) :

- **Libraries (like the ENSAM library) give you access to databases that are usually not free of charge :**
  - **Online scientific databases :** Science Direct, Springer Link, Scopus, etc.
  - **Encyclopaedia :** Techniques de l'ingénieur, etc.
  - **Patents and norms :** SagaWeb, etc.
  - **Open archives :** HAL, SAM, etc.
  - Etc.



- **Internet (the biggest single source of information in the world) :**
  - **Google scholar, ResearchGate :** to get access to scientific articles for free
  - **Data repository :** to complete your datasets
  - **Tools repository :** to benchmark your own approach and compare your results to the results of others.
  - Etc.



- **People :** Even if the world tends to be digitalized, **experts are still good means to access up-to-date and precise information** in a specific domain !



Loads of information

Quality rather than quantity (2/2)

The problem lies in where to find relevant information of the right quality !

The quality of the contents has to be tested :

- Is it accurate ? What are the sources ? Has the source been peer-reviewed ? Impact Factor of the journal ?
- What authority is it based on ? Are there recognized experts in that field ? Are there other references by the same authors ?
- Is it biased ? Who are the authors (university, league, company, influential group, etc.) ?
- How detailed is the information ? Is it self-contained or do you need to look for additional details ?
- Is it out of date ? When was it updated ?
- Is the information cross-checked with other sources ?





A four-hours training course is planned with Carole WAHNOUN to help you browsing the available sources.

Loads of information

Valorisation of research findings

The research findings are usually valorised following two main strategies :

- **Patent :** To protect an invention, usually when commercial matters are to be considered in a later stage. Very often considered when working with industrial partners.
- **Publication :** To spread the knowledge and make it accessible worldwide. Very often associated to Copyrights issues depending on the publisher.

The two strategies are not antinomic and a publication can follow a patent

The valorisation is often related to Intellectual Property (IP) issues

- **Definition :** intangible property that is the result of creativity
- **Protection :** by means of patents, copyrights, trademarks, etc.
- Usually requires the help of a lawyer to support the researcher and find the proper protection level.



It is important **to understand the rules associated to the sources** you refer to, in order to understand what you can do, and what you cannot do !

Also remember that IP issues may apply on your own project !

Types of publication

Most common types

Different kinds of paper (not exhaustive list) :

- **White paper :** It is an authoritative report or guide that informs readers concisely about a complex issue. It usually does not enter the details.
- **Technical paper :** It presents **findings associated to a particular research problem** in a given problem area. It contains a synthetic literature review (20-30 references), the details of the proposed methodology and methods forming the scientific contributions. Results show how the findings have been validated under specific hypotheses.
- **Survey paper :** It contains **a deep analysis of state-of-the-art approaches** to solve a well-identified research problem (>50 references). It contains a clear identification and definition of **the adopted criteria** as well as a deep analysis of existing approaches with respect to these criteria. It usually ends up with **clear identification of gaps and ways of improvement**.
- **Benchmark paper :** It contains a deep analysis of the behaviour of different approaches to solve the same problem. Approaches are discussed with respect to identified criteria used for the comparative study.

Doing a literature review

Four major directions

The literature review involves an objective critique and evaluation of the strengths and weaknesses of a document with respect to criteria



It should not be narrowly restricted to a specific subject area.

It can be carried out following 4 major directions, from a general to a particular level of analysis (Walliman, 2001) :

- **Research theory and philosophy :** to establish the intellectual context(s) of research related to your subject.
- **History of developments in your subject :** to trace the background to present thinking.
- **Latest research and developments in your subject** : to inform about the current issues being investigated and the latest thinking and practice, to discuss the conflicting arguments, and to detect a gap in knowledge.
- **Research methods** : to explore practical techniques that have been used, particularly those that might be relevant to your project.

Each document must be analysed to determine its quality and merits, and its relevance for your research topic.

Doing a literature review

Methodological aspects (1/2)

Checklist of questions to be considered during the literature review :

- What is the source of the article and year of publication ? As it been peer-reviewed ?
- Does the article state a research question and problem area ? Are the scientific contributions highlighted ? Are important notions introduced ? Are keywords suggested ?
- If applicable, does the article describe the data collection, or does it describe and synthesize other studies in which the data were collected ?
- Does the article include a section on literature review ? How deep is the analysis ? How relevant are the identified articles ? Is the literature review based on clearly identified and well formalized criteria ?
- Are the procedures followed clearly explained to allow the work to be repeated and the same results to be achieved ? If no, what is missing ?
- Do you agree with what has been done ? What more would you have done if you had done the research ?
- Does the article describe how the findings have been validated ? Do you agree with the interpretation of the results ? Why and why not ?
- What are the strengths and weaknesses ? Are the limitations clearly highlighted ? What could be reused in your own project ?

Doing a literature review

Methodological aspects (2/2)

Should you consider analysing the papers one by one, separately ?

- The literature review must be considered **as a whole** and documents must be compared the one with the others.
- Different levels can be associated to a particular criterion while considering **the relative efficiency** of the considered approaches. This requires having a good overview.

  Example : An algorithm can be considered as slow or fast (speed criterion) without necessarily relying on the use of timestamps.
- Most of the time, **absolute comparisons** can be hard to obtain, because of a lack of accessibility and openness of the methods developed by others. Ideally, all the methods should be accessible in order to allow for benchmarking.

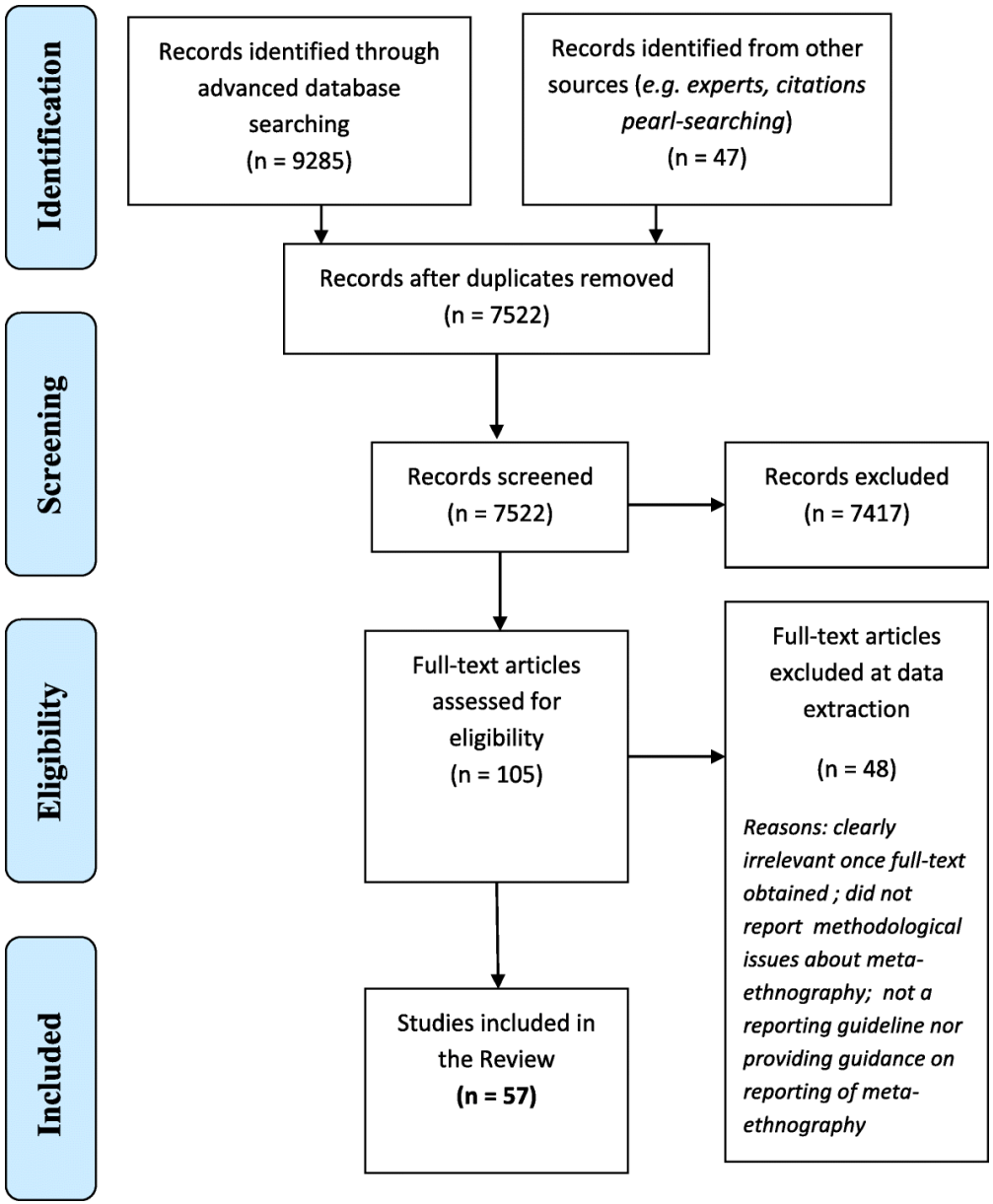
How many references do you need ? It depends but 15-20 references is a good trade-off for a master thesis, **if they are related to the research question !**

When should you stop ? Never, the literature review **is not a one-time step** ! Anyway, you can move on to the next step of the research process when you feel you are looping. But don't forget to come back on it later on.

Be organized and methodical. Make use of existing tools (Zotero, EndNote, etc.) to keep track of your readings and organize your analysis.

Doing a literature review

The PRISMA methodology



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Presenting your analysis

Key features of a literature review

References should be complete and easily accessible :

- Check that the name of the journal, the volume, the year, etc. are provided and presented in a uniform manner. Each reference must be cited in the core of the document.
- When considering an English-speaking audience, **avoid French references** and try to find the equivalent document in English.
- Do not rely on the same type of document. For instance, do not only cite PhD manuscript and **do prefer the published papers** that might have been written from the manuscript.

Your thoughts should be organized in a proper and intelligible way :

- Start describing the references as a whole and **try to identify categories of approaches** in order **to structure** your literature review. More general references are to be used to introduce the problem area and research question.
- **Define the criteria** used for the systematic analysis, as well as **the levels** associated to each criterion. There can be **qualitative and quantitative criteria**.
- **Analysis the documents** in light of the criteria. Some criteria may be non-applicable (n/a) or non-evaluable for some approaches and this must be clearly highlighted as well.
- Synthetize the overall analysis **by means of synthetic tables** where the reader can easily see the positioning of the approaches with respect to the adopted criteria.
- Clearly conclude by **stressing the gaps and ways of improvement**.

ETHICAL ISSUES

A two-level classification

Honesty in your work

Situations that raise ethical issues

Ethical issues

A two-level classification

Ethical issues can be considered at two levels (Walliman, 2011) :

- The **individual values of the researcher** relating to honesty, frankness and personal integrity when carrying on his/her research project.
- The **researcher's treatment of other people** involved in the research, relating to informed consent, confidentiality, anonymity and courtesy.

Both levels should be carefully considered. Not considering the ethical issues, and overstepping the rules, may lead to your research being discredited, and **to severe penalties and humiliation**.



Anyway, this is a brief introduction and if you have doubts during your project, **ask someone !**

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Honesty in your work

Most important issues

Ownership and plagiarism :

- Plagiarism happens **when directly copying someone else's work into your report**, thesis, etc. and **letting it be assumed that it is your own (i.e. without citing)** !
- Claiming sole authorship of work which is in fact the result of collaboration is also reprehensible to the law.

Acknowledgement and citation :

- Accusations of plagiarism can be avoided by acknowledging the sources and their originators within your own text. **This is called the citation**.
- No need to systematically used quote marks " " or italic typeface when referring to someone else work. This can be used with parsimony.
- More general **acknowledgements** can be written at the beginning or end of the document.

Responsibility of the researcher :

- The content should be sufficiently detailed, argued and **limitations clearly highlighted**.
- **Distorting your data or results** knowingly is a serious lapse of honesty. Silently rejecting or ignoring evidence, being too selective in the data used and in presenting the results constitutes a breach of integrity. **Possible bias in any aspect of the research are to be acknowledged and explained**.

Situations that raise ethical issues

Some hints

Research aim : it must comply with the University's requirements and its ethical rules…

Language : the way you present and discuss your research project should be as neutral as possible. No patronizing or disparaging, avoid bias, stereotyping, discrimination, prejudice, intolerance and discrimination…

Presentation : when presenting yourself (for an interview in the street or elsewhere), you should present who you are and not overestimating yourself…

Dealing with participants : this is a hot topic with respect to ethical issues and it has to be treated very carefully. You have to comply to the rules of your University, funding structure, country, etc… Participants notably have to notified on what the data will be used for and they do have to give an **informed consent**…

Data treatment and storing : data must be treated very carefully to avoid distortions, modifications, simplifications etc. of the raw data…

Data protection : the researcher must comply with the rules existing in his/her country, state etc. and let the possibility the modification of the participants' data…

Ex: RGPD (Règlement Général sur la Protection des Données) in France.

…

CONCLUSION

Next steps

Conclusion
Next steps

Multiple notions related to research methodology have been introduced :

- Scientific method, its assumptions and key characteristics
- Different types of research and their objectives
- Problem area and research problem identification
- Research process and its desirable characteristics
- Difference between method and methodology
- How the research problem is to be identified
- The literature review and its methodological aspects
- Ethical issues

Additional references have been proposed to complete this introduction



These multiple aspects of the **early stages of the research process** will be first illustrated during the PJR (128h) and further experienced during the full-time master thesis (6 months).